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LESSON 1: Library management system

- Identify candidate classes (Books, Readers, Vouchers, Employees, Publishers)

+ Book: main entity, managing book information.

+ DocGia: person who borrows books.

+ PhieuMuon: records borrowing/returning.

+ NhanVien: system manager (librarian, warehouse keeper...).

+ NhaXuatBan: book source information.

- Properties and behavior (methods) of each class

+ Sach

o Attributes: maSach (PK, string), nameSach (string), tacGia (string),

namXuatBan (int), maNXB (FK), soLuongTon (int), trangThai (string: "Available"/"On loan").

o Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(),
capNhatTonKho ().

+ DocGia

o Attributes: maDocGia (PK, string), hoTen (string), birth date (date), diaChi (string), soDienThoai (string), ngayDangKy (date), gioiTinh (boolean).

o Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon().

+ PhieuMuon

o Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK),

maNhanVien (FK), nayMuon (date), nayHenTra (date), nayTraThucTe (date), TinhTrang (string: "Borrowing"/"Returned"/"Overdue").

o Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(),

crystalPhieuPhat(). + NhanVien

o Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), ngaySinh (date), soDienThoai (string).

o Behavior: dangNhap(), quanLyDocGia(), quanLyMuonTra(), quanLySach ().

+ NhaXuatBan

o Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email (string).

o Behavior: addNXB(), capNhatNXB().

- Relationships

+Books and Publishers: 1 book – 1 publisher

- ++Readers and Borrowing Slips: 1 reader – many borrowing slips
- ++Books and loan slips: 1 book - many loan slips
- ++Employees and Loan Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- + DangKyTaiKhan (Customer registration).
- + DangNhap (Sign in).
- + SeeSanPham (Search / View product list).
- + ThemVaoGioHang (Add to cart).
- + DatHang (Order).
- + ThanhToan (Online payment / COD).
- + View DonHang (View order history).

Admin / NguoiQuanLy:

- + QuanLySanPham (Add/Edit/Delete products).
- + QuanLyDonHang (View/Confirm/Cancel order).
- + QuanLyKhachHang (Customer information management).
- + QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- + ThanhToan includes XacNhanDonHang.
- ++DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- ++ Customers (Customer): from register, view orders.
- ++SanPham (Product): product management.
- ++GioHang (Cart): temporarily save the product before ordering.
- ++ Don Hang (Order): from order, payment.
- ++ChiTietDonHang (OrderDetail): product details in the order.
- ++ ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- ++Admin or NguoiQuanLy (inherits from User if there is a user system).

- Analyze relationships between classes (specify inheritance if any)
- +KhachHang and DonHang: 1 customer – many orders
- +DonHang and ChiTietDonHang: 1 order – many order details (Composition)
- +ChiTietDonHang and SanPham: many single details – 1 product
- +DonHang and ThanhToan: 1 order – 1 payment
- +User (abstract class) ←inherit ←KhachHang and Admin (KhachHang and Admin are both a type of User, sharing the same account, password, full name).

Lesson 3: Convert class diagrams into data tables

1.MonHoc Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	